

Supplementary File S1 — TRIPOD 2015 Checklist

Manuscript: Beyond Binary Lock: A Continuous Aperture-Collapse Hazard Model of Middle-Ear Barotrauma in Hypobaric-Chamber Training

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Reporting guideline: Collins GS, Reitsma JB, Altman DG, Moons KGM. Transparent Reporting of a multivariable prediction model for Individual Prognosis Or Diagnosis (TRIPOD): The TRIPOD Statement. *Ann Intern Med.* 2015;162(1):55–63. doi:10.7326/M14-0697.

Study type (TRIPOD Type): Type 2a — development of a prediction model with external validation using separately collected data from the original dataset population. Development anchor: Colombian Aerospace Force (FAC) DIMAE 2010–2026 cohort. External validation: three previously published independent military-aviation cohorts (Morgagni 2010; Morgagni 2012; Landolfi 2009).

TRIPOD 2015 Checklist (22 items)

#	Section / topic	Checklist item	Development	Validation	Page / section in manuscript
1	Title	Identify the study as developing and/or validating a multivariable prediction model, the target population, and the outcome to be predicted.	Yes	Yes	Title, p. 1 — “Beyond Binary Lock: A Continuous Aperture-Collapse Hazard Model of Middle-Ear Barotrauma in Hypobaric-Chamber Training” — explicitly identifies the prediction model and the target outcome (middle-ear barotrauma) in the chamber-training population.
2	Abstract	Provide a summary of objectives, study design, setting, participants, sample size, predictors, outcome, statistical analysis, results, and conclusions.	Yes	Yes	§Abstract — Background / Objective / Methods / Results / Conclusion.

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3a	Introduction — Background and objectives	Explain the medical context (including whether diagnostic or prognostic) and rationale for developing or validating the multivariable prediction model, including references to existing models.	Yes	Yes	§1 Introduction. Kanick-Doyle 2005 reference model and three stated shortcomings.
3b	Introduction — Background and objectives	Specify the objectives, including whether the study describes the development or validation of the model, or both.	Yes	Yes	§1, final paragraph: "extension that addresses these gaps, calibrated to the Colombian Aerospace Force (FAC) 10-year operational registry and externally validated on three independent previously published military-aviation cohorts."
4a	Methods — Source of data	Describe the study design or source of data (e.g., randomized trial, cohort, or registry data), separately for the development and validation data sets, if applicable.	Yes	Yes	§2.8 "Source of data and participants (development cohort)"; §2.8 "External validation."
4b	Methods — Source of data	Specify the key study dates, including start of accrual; end of accrual; and, if applicable, end of follow-up.	Yes	Yes	§2.8: "1 January 2010 and 31 March 2026 (n = 7,271 exposures)." External-validation cohorts cited with original publication years.
5a	Methods — Participants	Specify key elements of the study setting (e.g., primary care, secondary care, general population) including number and location of centres.	Yes	Yes	§2.8: "DIMAE hypobaric chamber facility (Bogotá, 2,640 m elevation)." External-validation cohorts per original publications.

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5b	Methods — Participants	Describe eligibility criteria for participants.	Yes	Yes	§2.8: “active-duty and reserve military aviators and aircrew of the Colombian Aerospace Force... no exclusion by age, sex, or prior ear history.”
5c	Methods — Participants	Give details of treatments received, if relevant.	Yes	Yes	Not applicable — no therapeutic intervention is modelled; model predicts MEB risk conditional on a chamber profile.
6a	Methods — Outcome	Clearly define the outcome that is predicted by the prediction model, including how and when assessed.	Yes	Yes	§2.8 “Outcome.” Clinically diagnosed middle-ear barotrauma (Teed grade $\geq$ I) within 24 h of exposure, ascertained by chamber flight surgeon.
6b	Methods — Outcome	Report any actions to blind assessment of the outcome to be predicted.	Yes	Yes	Not applicable — clinical outcome ascertainment was part of routine post-exposure medical review; no blinded adjudication panel was used. Stated as a limitation in §4.1.
7a	Methods — Predictors	Clearly define all predictors used in developing the multivariable prediction model, including how and when they were measured.	Yes	Yes	§2.3 Physics core; §2.4 Aperture-collapse; §2.5 Pathophysiology state machine; §2.8 “Predictors” list. Table I lists URI predictor multipliers.
7b	Methods — Predictors	Report any actions to blind assessment of predictors for the outcome and other predictors.	Yes	Yes	Not applicable — predictors are mechanistic model parameters and state variables, not clinician-rated fields.

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8	Methods — Sample size	Explain how the study size was arrived at.	Yes	Yes	§2.8 Development n = 7,271 exposures (pooled registry, all available). External-validation n per original publications (Morgagni 2010 n = 1,241; Morgagni 2012 n = 314; Landolfi 2009 n = 335).
9	Methods — Missing data	Describe how missing data were handled (e.g., complete-case analysis, single imputation, multiple imputation) with details of any imputation method.	Yes	Yes	§2.8 “Missing data.” No imputation required — calibration anchor is an aggregate event rate.
10a	Methods — Statistical analysis	Describe how predictors were handled in the analyses.	Yes	Yes	§2.3–§2.7 describe mechanistic handling. §2.8 “Predictors”: no variable selection; all retained by mechanistic construction.
10b	Methods — Statistical analysis	Specify type of model, all model-building procedures (including any predictor selection), and method for internal validation.	Yes	Yes	§2.7 Three-threshold cumulative hazard; §2.8 “Calibration methods” (log-space bisection and ABC-SMC). Internal validation by pinned Kanick-Doyle/Groth 1986 regression fixture (§3.4, §2.9).
10c	Methods — Statistical analysis (validation)	For validation, describe how the predictions were calculated.	—	Yes	§2.8 “External validation”: simulator run on matched reconstructed chamber profile with tightened cohort prior reflecting tympanometry pre-screening reported in the original publications; no refitting of hazard constants.

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10d	Methods — Statistical analysis	Specify all measures used to assess model performance and, if relevant, to compare multiple models.	Yes	Yes	\$2.8 “Model performance measures” — CI coverage, career-3 projection agreement, pinned Kanick-Doyle fixture parity.
10e	Methods — Statistical analysis	Describe any model updating (e.g., recalibration) arising from the validation, if done.	—	Yes	Not done — external validation reports transfer performance without refitting. Stated explicitly in \$2.8 “External validation” and \$4.3 Conclusions.
11	Methods — Risk groups	Provide details on how risk groups were created, if done.	Yes	Yes	Not applicable — model output is continuous per-exposure probabilities; risk categorisation (low / moderate / high / very_high) is a derived label only in the user-facing API (documented in the model card), not used in the analysis or in validation.
12	Methods — Development vs. validation	For validation, identify any differences from the development data in setting, eligibility criteria, outcome, and predictors.	—	Yes	\$2.8 External-validation paragraph; \$4.1 Limitations (external-validation cohorts used tympanometry pre-screening; FAC cohort did not).

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13a	Results — Participants	Describe the flow of participants through the study, including the number of participants with and without the outcome and, if applicable, a summary of the follow-up time. A diagram may be helpful.	Yes	Yes	§3.1 Calibration (173/7,271 events, URI subgroup means). §3.2 External validation. No flow diagram is provided as participants are not individually tracked — aggregate denominators are reported.
13b	Results — Participants	Describe the characteristics of the participants (basic demographics, clinical features, available predictors), including the number of participants with missing data for predictors and outcome.	Yes	Yes	§2.8 “Source of data and participants.” Individual-level demographics not reported due to institutional and personal data-protection constraints (§Declarations — Data availability).
13c	Results — Participants (validation)	For validation, show a comparison with the development data of the distribution of important variables (demographics, predictors, and outcome).	—	Yes	§3.2 and Table III: observed prevalence and Wilson 95% CIs per cohort. External-validation cohort per-exposure rates (1.5–2.5%) are within the same order as the pooled FAC 2.38%, as discussed in §4.1 Limitations.
14a	Results — Model development	Specify the number of participants and outcome events in each analysis.	Yes	—	§2.8 (n = 7,271 exposures, 173 events); §3.1 Calibration.

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14b	Results — Model development	If done, report the unadjusted association between each candidate predictor and outcome.	Yes	—	Not applicable — the model is mechanistic with pre-specified predictor structure; no regression-style unadjusted associations are estimated. URI subgroup event rates in §3.1 serve as the closest analogue (stratified means).
15a	Results — Model specification	Present the full prediction model to allow predictions for individuals (i.e., all regression coefficients, and model intercept or baseline survival at a given time point).	Yes	—	§2.3–§2.7 specify all model equations and parameter values; Table II reports calibrated hazard constants ($r_{\text{barotitis}} = 5.85 \times 10^{-8}$, $r_{\text{baromyringitis}} = 1.75 \times 10^{-9}$, $r_{\text{rupture}} = 5.85 \times 10^{-11}$). Open-source code at the repository reproduces the full specification.
15b	Results — Model specification	Explain how to use the prediction model.	Yes	—	Model card §2–§4 (instruction to supply a <code>PatientState</code> and a <code>ChamberProfile</code> ; outputs per-exposure probabilities with optional credible intervals).
16	Results — Model performance	Report performance measures (with CIs) for the prediction model.	Yes	Yes	§3.1 Calibration: simulated 2.47% per-exposure vs. target 2.38%; career-3 7.23% vs. 6.97%. §3.2 External validation with point estimates and observed Wilson 95% CIs (Table III).

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17	Results — Model-updating	If done, report the results from any model updating (i.e., model specification, model performance).	—	Yes	Not done; see item 10e.
18	Discussion — Limitations	Discuss any limitations of the study (such as non-representative sample, few events per predictor, missing data).	Yes	Yes	§4.1 Limitations — unpublished anchor cohort, conservative rupture threshold, explicit modeler priors, no per-subject ML head, approximated muscle mechanics, reconstructed validation chamber profiles.
19a	Discussion — Interpretation (validation)	For validation, discuss the results with reference to performance in the development data, and any other validation data.	—	Yes	§4 Discussion and §4.1 Limitations.
19b	Discussion — Interpretation	Give an overall interpretation of the results, considering objectives, limitations, results from similar studies, and other relevant evidence.	Yes	Yes	§4 Discussion (three extensions deserving comment); §4.2 Incorporation of 2025–2026 evidence; §4.3 Conclusions.
20	Discussion — Implications	Discuss the potential clinical use of the model and implications for future research.	Yes	Yes	§4.3 Conclusions — clinical use (profile-risk ranking, URI dominance as modifiable risk driver) and future work (FAC per-TEED-grade breakdown; aperture half-point empirical refinement). Model card §7 enumerates known risks of misuse.

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21	Other — Supplementary information	Provide information about the availability of supplementary resources, such as study protocol, Web calculator, and data sets.	Yes	Yes	§Declarations — Code availability (GitHub repository, MIT license, v2.2.1-manuscript tag, Zenodo DOI planned); Data availability; Model card provided in repository.
22	Other — Funding	Give the source of funding and the role of the funders for the present study.	Yes	Yes	§Declarations — Funding. No external funding; DIMAE institutional resources only.

Summary

All 22 TRIPOD 2015 items are addressed in the manuscript or in the accompanying open-source model card, with explicit page / section pointers above. Items 10e, 17 (model-updating) and 14b (unadjusted predictor-outcome associations) are not applicable to a mechanistic physics-informed model; these are reported as “Not done” or “Not applicable” with a rationale.

TRIPOD+AI (Collins 2024) does not apply: no machine-learning component is used in the deployed prediction pipeline (see §2.1 Study design and Declarations — Reporting guideline).